Comprehensive Report: MNIST Clustering and Dimensionality Reduction

Link to the notebook

Executive Summary

This notebook explored the MNIST handwritten-digit dataset using unsupervised learning and dimensionality-reduction techniques. The workflow normalized the 784 pixel features, examined the class distribution, clustered a manageable subset with K-Means, evaluated the resulting clusters with inertia, silhouette score, and label composition, and visualized the structure with PCA and t-SNE.

The analysis demonstrates that K-Means can organize handwritten digits into meaningful groups, but the clusters do not necessarily correspond one-to-one with the ten known digit labels. PCA provides a fast global projection, while t-SNE is more effective for inspecting local neighborhoods and visual separation. These methods are complementary: neither two-dimensional visualization nor cluster IDs should be interpreted as a definitive digit classifier.

1. Objective and Questions

The analysis addressed the following questions:

1. How should MNIST pixel data be prepared for clustering?
2. How many clusters provide a reasonable representation of the digit data?
3. How well do the unsupervised clusters align with the known digit labels?
4. Can the trained clustering model assign groups to previously unused images?
5. What structure is revealed by PCA and t-SNE projections?

The true labels were retained for analysis and validation only. They were not supplied to K-Means during training.

2. Data Preparation and Exploratory Analysis

MNIST contains grayscale images of handwritten digits. Each image is represented as a flattened vector of 784 pixel values, corresponding to a 28 x 28 image. The data was loaded from OpenML with `fetch_openml` and split conceptually into features (`X`) and reference labels (`y`).

Pixel values were divided by 255 so that every feature lies on an approximately 0-to-1 scale. This normalization prevents the magnitude of raw 8-bit pixel values from dominating Euclidean distance calculations used by K-Means and the dimensionality-reduction methods.

A count plot of `y` was used as a basic data-quality and balance check. It provides context for interpreting cluster composition: a cluster containing several labels may reflect genuine visual similarity, while an apparent imbalance may partly reflect the distribution of examples in the selected data.

## 3. Clustering Methodology

Because the complete dataset contains 70,000 observations and 784 features, the notebook used the first 10,000 normalized observations for the main clustering exercise. This decision reduces runtime and memory requirements while preserving a sufficiently large sample for demonstration and visualization.

K-Means was evaluated for values of `K` from 1 through 20 using within-cluster sum of squares, also called WCSS or inertia. The elbow plot was used as a diagnostic for identifying where additional clusters begin to yield diminishing reductions in WCSS. Ten clusters were then selected because MNIST contains ten digit classes and because this value offers a natural, interpretable comparison with the known labels.

The final model used:

- `n_clusters=10`
- `random_state=42` for reproducibility
- `n_init=10` to reduce sensitivity to initialization
- `max_iter=300` and `tol=1e-4` for convergence control

The model was fit only to `X_subset`, and the resulting assignments were stored in `cluster_labels`.

## 4. Cluster Analysis and Findings

The centroid images show the average pixel pattern for each K-Means cluster. Centroids that resemble recognizable digits indicate groups with relatively consistent visual structure. Less distinct centroids suggest that a cluster combines multiple writing styles or overlapping digit shapes.

The cluster-versus-label count plot and the crosstab heatmap provide a more reliable interpretation than centroid appearance alone. They show which true labels are concentrated in each cluster and where label mixing occurs. Because K-Means optimizes geometric compactness rather than label purity, mixed clusters are expected, especially for visually similar digits such as 3/5, 4/9, or 1/7. Cluster IDs are arbitrary identifiers and should not be interpreted as digit names.

The silhouette score summarizes cohesion and separation for the selected clustering. A higher value indicates that observations are generally closer to their assigned cluster than to neighboring clusters. A moderate or low value would be consistent with overlapping handwriting styles and the difficulty of representing complex digit manifolds with spherical, Euclidean K-Means clusters. The exact score should be taken from the printed result in the preceding cell.

5. Prediction on Unseen Data

The K-Means model was applied to a small slice beginning at `n_samples_subset`, ensuring that these images were not used during fitting. The predicted cluster IDs were displayed alongside the images, and the original labels were printed for comparison.

This step confirms that the fitted model can assign new observations to the nearest learned centroid. It is an assignment demonstration rather than a supervised accuracy test: the predicted cluster number has no inherent relationship to the digit label, so cluster-to-digit mapping would need to be learned separately before calculating classification accuracy.

6. Dimensionality Reduction

PCA

PCA reduced the 784-dimensional normalized observations to two components. The PCA scatter plot provides a global, linear summary of the largest directions of variance. It is computationally efficient and useful for identifying broad trends, but distinct digit groups may overlap because two components cannot preserve all of the original information.

t-SNE

t-SNE was applied to a reproducible random sample of 2,000 observations from the clustering subset. The smaller sample was chosen because t-SNE is substantially more computationally expensive than PCA. With `random_state=42`, `perplexity=30`, and `max_iter=1000`, the resulting projection emphasizes local neighborhoods and often produces visually separated islands.

The t-SNE plot can make clusters easier to inspect visually, but distances between distant islands, cluster sizes, and apparent global geometry should not be overinterpreted. t-SNE is primarily a visualization tool, not a clustering objective or a proof that the underlying classes are linearly separable.

## 7. Overall Insights

- Normalization was essential because the algorithms rely on distances and variance.
- The 10,000-row subset made the analysis practical while preserving the central workflow.
- The choice of ten clusters is interpretable for MNIST, but the elbow plot and silhouette score remain important empirical diagnostics.
- K-Means captures broad visual groupings, yet handwriting variation causes overlap and mixed-label clusters.
- Centroids, label-composition tables, and evaluation metrics should be interpreted together.
- PCA is best suited to a fast global overview; t-SNE is better suited to local-neighborhood visualization.
- Unsupervised cluster IDs cannot be treated as predicted digit labels without an additional mapping step.

## 8. Limitations and Recommended Extensions

This notebook uses only the first 10,000 observations for clustering and a 2,000-observation sample for t-SNE, so the findings are representative of the selected data rather than a full-dataset guarantee. K-Means also assumes compact clusters under Euclidean distance, which is a simplified model for handwritten-image structure. The silhouette score is computed on the training subset and therefore does not measure generalization to a separate test set.

Useful extensions would include comparing multiple random subsets, using `MiniBatchKMeans` for the full dataset, evaluating several values of `K` with both silhouette score and label-alignment measures, and comparing alternative methods such as Gaussian mixture models or spectral clustering. For supervised digit recognition, a classifier trained on the labels would be more appropriate than interpreting K-Means cluster IDs directly.

## 9. Reproducibility

The main stochastic operations use `random_state=42` or `np.random.seed(42)`. Re-running the cells in order reproduces the preprocessing, subset selection, K-Means initialization behavior, and t-SNE sample selection. The OpenML download requires network access and a working scikit-learn installation. Numerical outputs should be read from the executed cells because runtime, library versions, and dataset availability can affect exact values and timings.

## Conclusion

The notebook presents an end-to-end unsupervised analysis of MNIST: preparation, exploratory inspection, K-Means clustering, cluster validation, prediction on held-out examples, and two-dimensional visualization. The central finding is that dimensionality reduction helps reveal structure, while cluster composition and quantitative metrics are needed to judge whether that structure corresponds to the known digit categories. Together, the results provide a practical demonstration of both the usefulness and the limits of unsupervised clustering for high-dimensional image data.